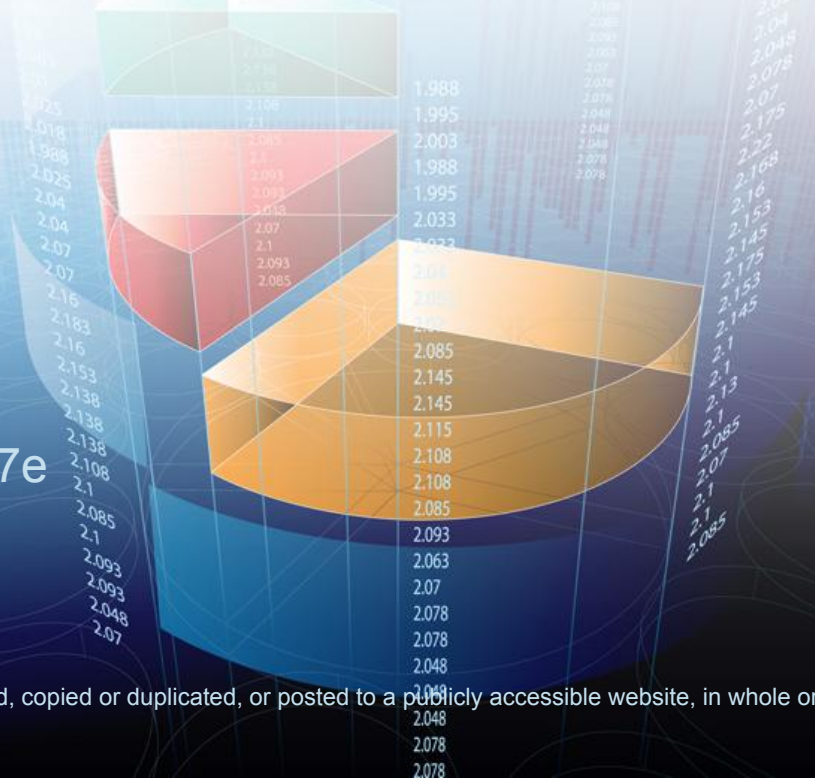
# Chapter 7
# The Conversion Cycle



Accounting Information Systems, 7e
James A. Hall



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# Objectives for Chapter 7

- Elements and procedures of a traditional production process
- Data flows and procedures in a traditional cost accounting system
- Accounting controls in a traditional environment
- Principles, operating features, and technologies of lean manufacturing
- Shortcomings of traditional accounting methods in the world-class environment
- Key features of activity based costing and value stream accounting
- Information systems of lean manufacturing and world-class companies

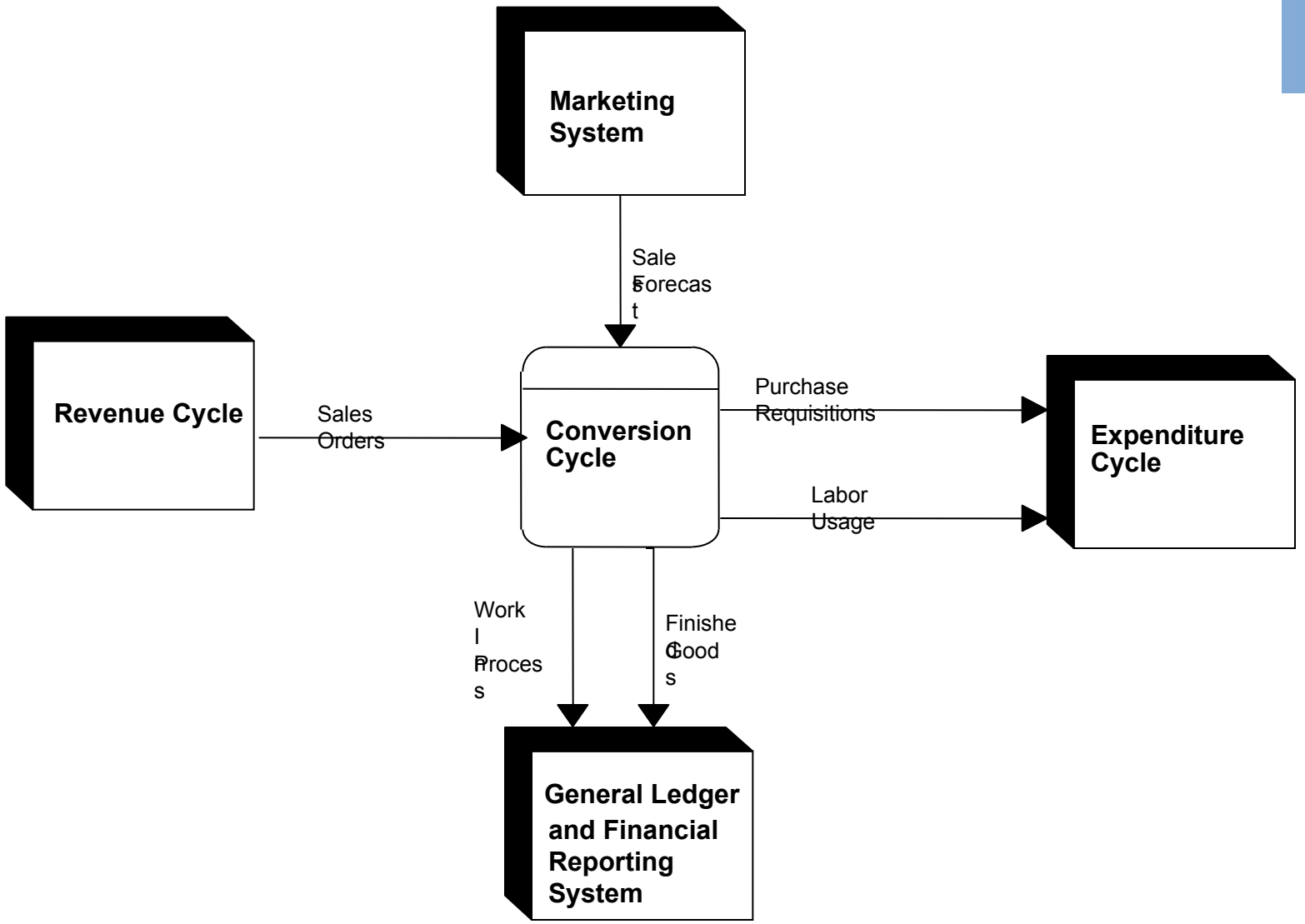

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# The Conversion Cycle

- Transforms input resources, raw materials, labor, and overhead into finished products or services for sale
- Consists of two subsystems:
  - **Physical activities – the production system**
  - **Information activities – the cost accounting system**


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# Conversion Cycle in Relation to Other Cycles





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# Production System

- Involves the planning, scheduling, and control of the physical product through the manufacturing process
  - determining raw materials requirements
  - authorizing the release of raw materials into production
  - authorizing work to be conducted in the production process
  - directing the movement of work through the various stages of production


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# Production Methods

- **Continuous Processing** creates a homogeneous product through a continuous series of standard procedures.
- **Batch Processing** produces discrete groups (batches) of products.
- **Make-to-Order Processing** involves the fabrication of discrete products in accordance with customer specifications.


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# Overview: Traditional Batch Production Model…

- consists of four basic processes:
  - plan and control production
  - perform production operations
  - maintain inventory control
  - perform cost accounting


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# Batch Production System

- ***Production Planning and Control***
  - Materials and operations requirements
  - Production scheduling
- ***Materials and Operations Requirements***
  - *Materials requirement* – the difference between what is needed and what is available in inventory
  - *Operations requirements* – the assembly and/or manufacturing activities to be applied to the product


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# Batch Production System

- ***Production Scheduling***
  - Coordinates the production of multiple batches
  - Influenced by time constraints, batch size, and other specifications
- ***Work Centers and Storekeeping***
  - Production operations begin when work centers obtain raw materials from *storekeeping.*
  - It ends with the completed product being sent to the *finished goods (FG) warehouse .*


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# Batch Production System

- ***Inventory Control***
  - *Objective:* minimize total inventory cost while ensuring that adequate inventories exist of production demand
  - Provides production planning and control with status of finished goods and raw materials inventory
  - Continually updates the raw material inventory during production process
  - Upon completion of production, updates finished goods inventory

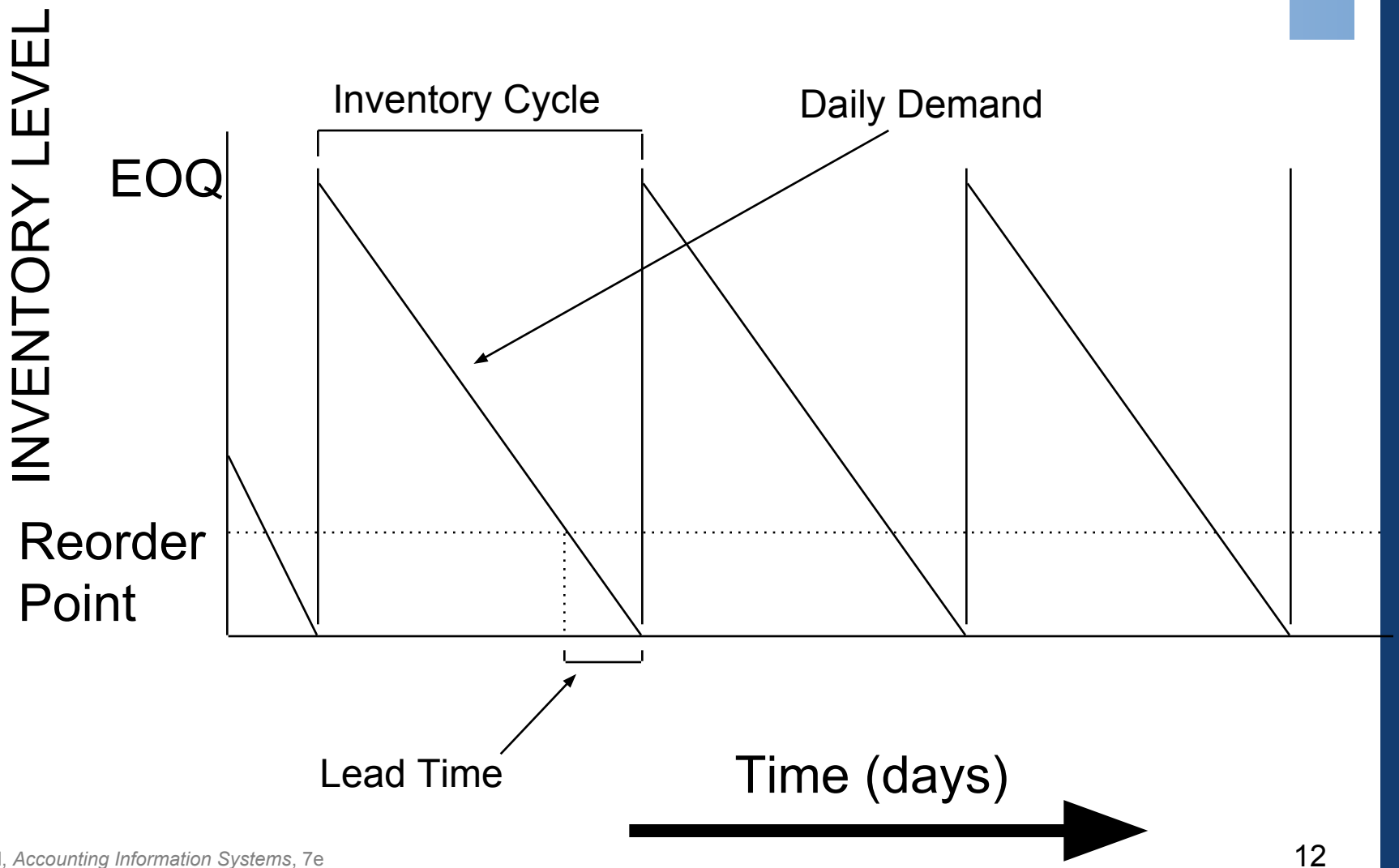

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# EOQ Inventory Model

- Very simple too use, but assumptions are *not always valid*
  - demand is known and constant
  - ordering lead time is known and constant
  - total cost per year of placing orders decreases as the order quantities increase
  - carrying costs of inventory increases as quantity of orders increases
  - no quantity discounts


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# EOQ Inventory Model





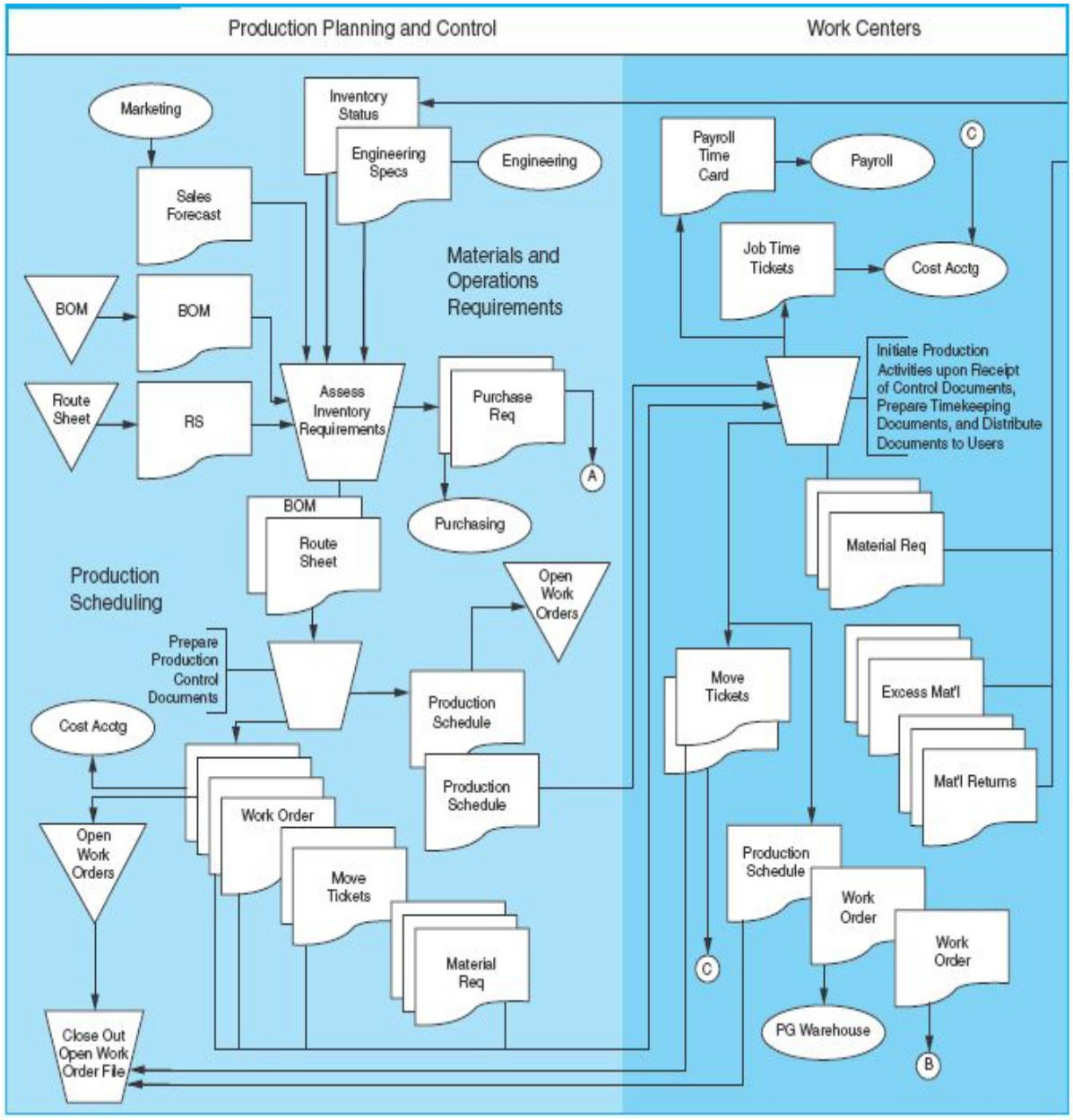
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# Information: Documents in the Batch Production System

- **Sales Forecast -** expected demand for the finished goods
- **Production Schedule** - production plan and authorization to produce
- **Bill of Materials (BOM)** - specifies the types and quantities of the raw materials and subassemblies used to produce a single finished good unit



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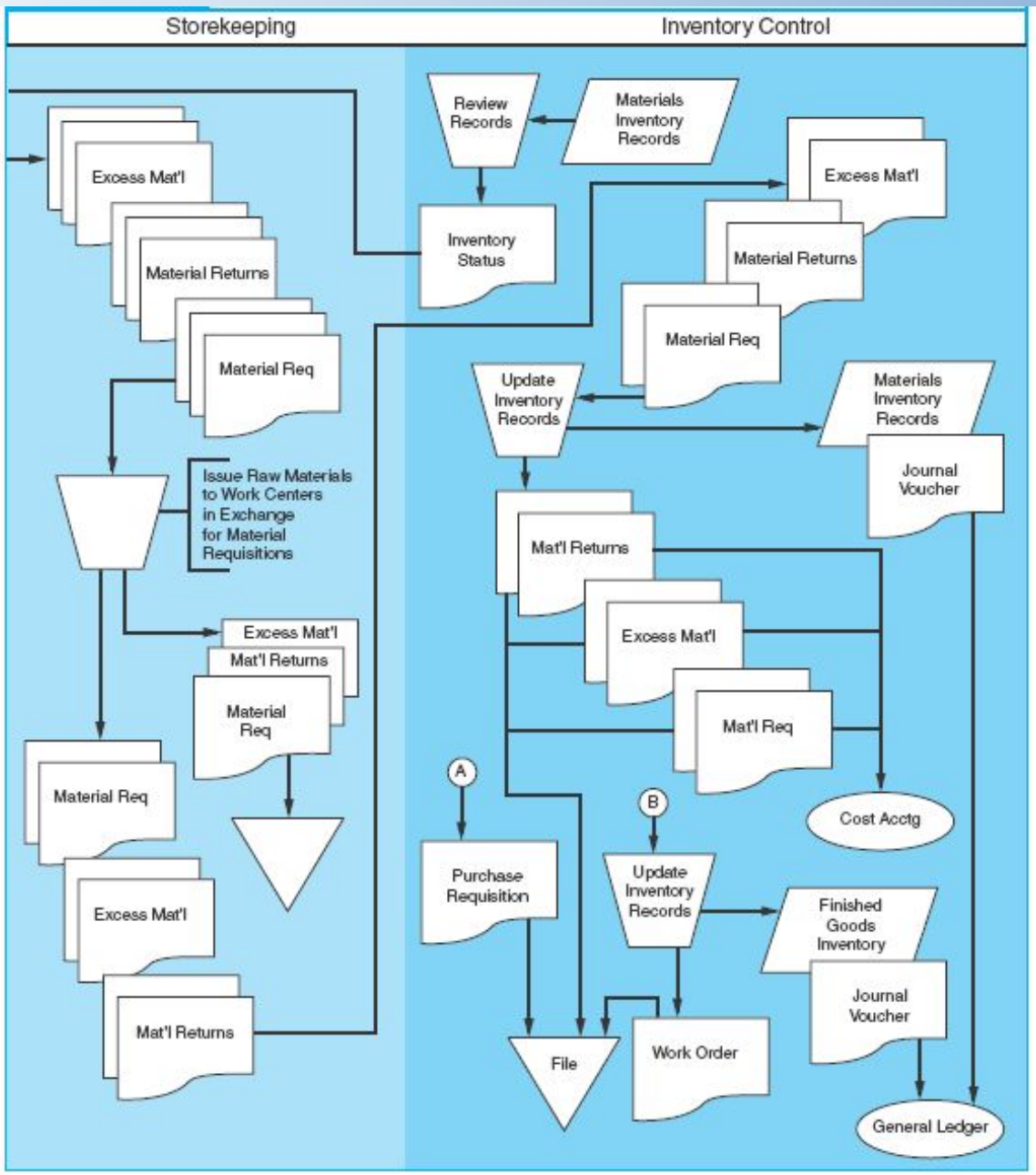



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# Information: Documents in the Batch Production System

- **Route Sheet** - details the production path a particular batch will take in the manufacturing process
  - sequence of operations
  - time allotted at each station
- **Work Order** - uses the BOM and route sheet to specify the exact materials and production processes for each batch


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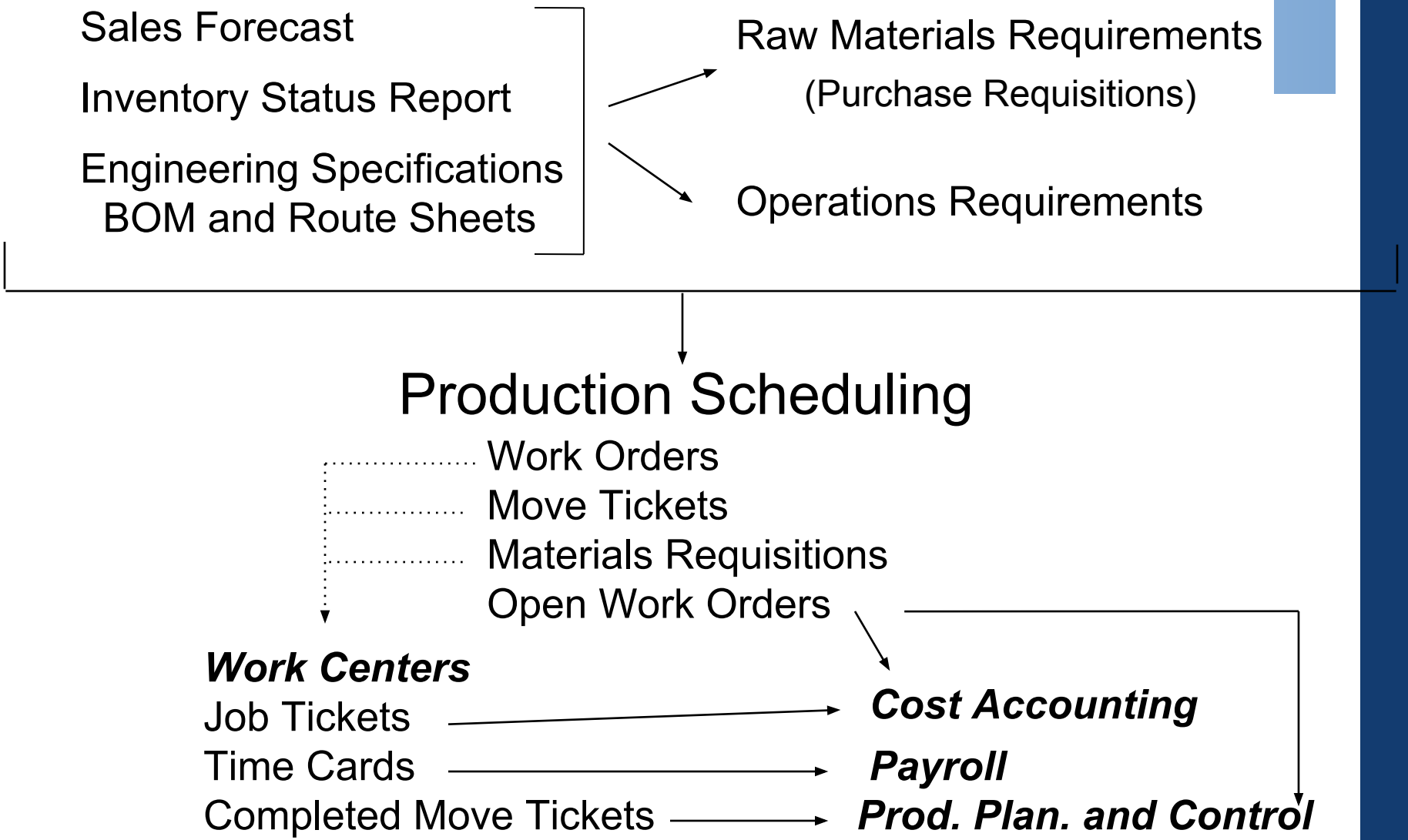
# Information: Documents in the Batch Production System

- **Move Ticket** - records work done in each work center and authorizes the movement of the batch
- **Materials Requisition** - authorizes the inventory warehouse to release raw materials for use in the production process



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# Production Planning and Control






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# Upon Completion of the Production Process…

Finished Product and Closed Work Order → ***Finished Goods Warehouse***

Closed Work Order → ***Inventory Control***

Status Report of Raw Materials and Finished Goods → ***Prod. Plan. and Control***

Journal Voucher → ***General Ledger***



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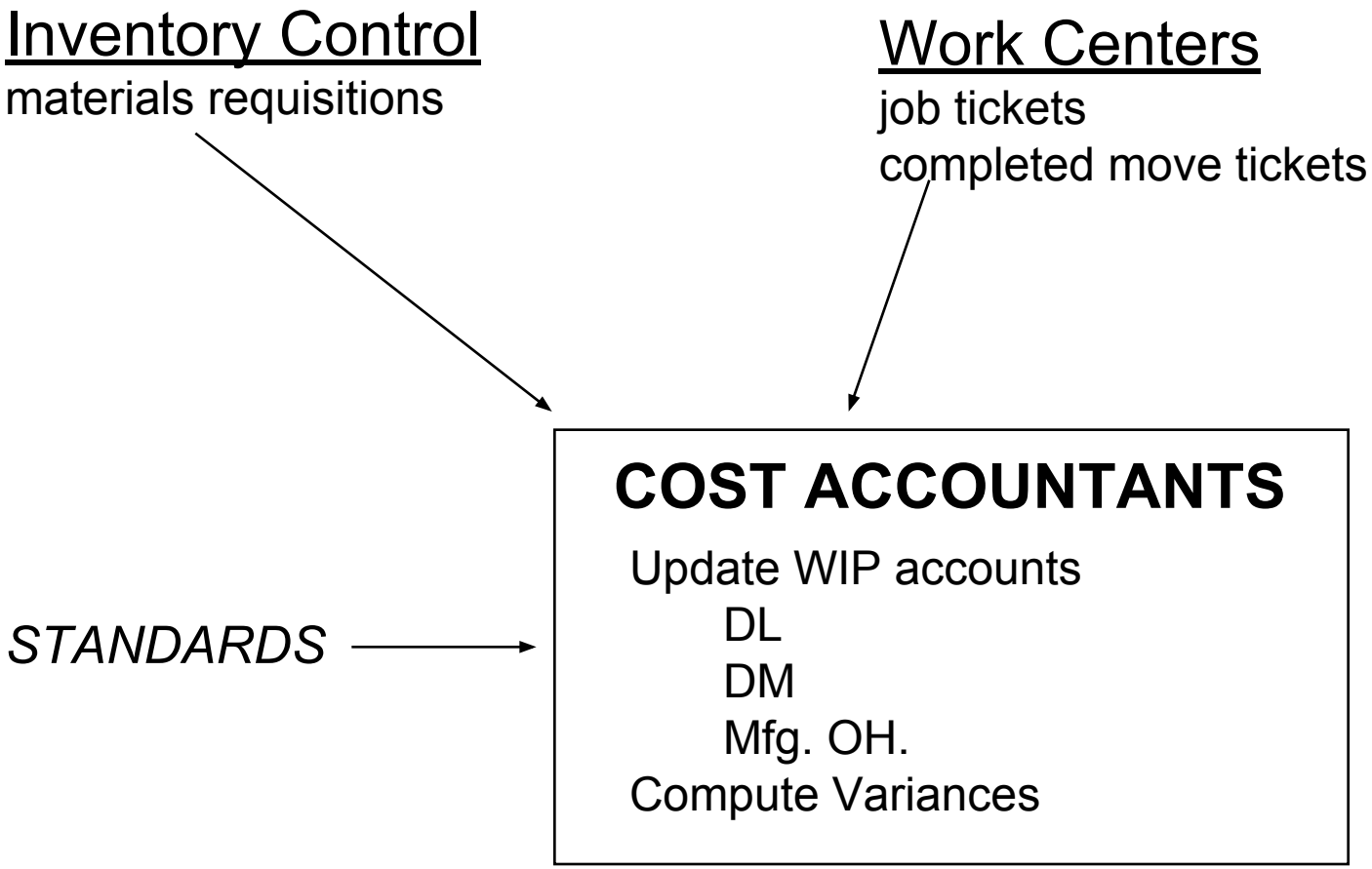
# Cost Accounting System

- Records the financial effects of the events occurring in the production process
- Initiated by the work order
- Cost accounting clerk creates a new cost record for the batch and files in WIP file
- The records are updated as materials and labor are used



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# Elements of the Cost Accounting System





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# Cost Accounting System

- Receipt of last move ticket signals completion of the production process
  - clerk removes the cost sheet from WIP file
  - prepares a journal voucher to transfer balance to a finished goods inventory account  and forwards to the General Ledger department



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# Summary of Internal Controls

| Control Class | Control Points in the System |
|---|---|
| Transaction authorization | Work orders, move tickets, and materials requisitions. |
| Segregation of duties | 1. Inventory control separate from RM and FG inventory custody.<br>2. Cost accounting separate from work centers.<br>3. GL separate from other accounting functions. |
| Supervision | Supervisors oversee usage of raw materials and timekeeping. |
| Access | Limit physical access to finished goods, raw materials stocks, and production processes. Use formal procedures and documents to release materials into production. |
| Accounting records | Work orders, cost sheets, move tickets, job tickets, materials requisitions, WIP records, FG inventory file. |
| Independent verification | Cost accounting function reconciles all cost of production.<br>General ledger reconciles overall system. |


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# Internal Controls

- Transaction authorizations
  - ***work orders*** **–** reflect a legitimate need based on sales forecast and the finished goods on hand
  - ***move tickets*** **–** signatures from each work station authorize the movement of the batch through the work centers
  - ***materials requisitions*** **–** authorize the warehouse to release materials to the work centers


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# Internal Controls

- Segregation of duties
  - production planning and control department is separate from the work centers
  - inventory control is separate from materials storeroom and finished goods warehouse
  - cost accounting function accounts for WIP and should be separate from the work centers in the production process



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# Internal Controls

- Supervision
  - work center supervisors oversee the usage of raw materials to ensure that all released materials are used in production and waste is minimized
  - employee time cards and job tickets are checked for accuracy



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# Internal Controls

- Access control
  - **direct** access to assets
    - controlled access to storerooms, production work centers, and finished goods warehouses
    - quantities in excess of standard amounts require approval
  - **indirect** access to assets
    - controlled use of materials requisitions, excess materials requisitions, and employee time cards


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# Internal Controls

- Accounting records
  - pre-numbered documents
  - work orders
  - cost sheets
  - move tickets
  - job tickets
  - material requisitions
  - WIP and finished goods files



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# Internal Controls

- Independent verification
  - cost accounting reconciles material usage (material requisitions) and labor usage (job tickets) with standards
    - variances are investigated
  - GL dept. verifies movement from WIP to FG by reconciling journal vouchers from cost accounting and inventory subsidiary ledgers from inventory control
  - internal and external auditors periodically verify the raw materials and FGs inventories through a physical count



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# World-Class Companies…

- continuously pursue improvements in all aspects of their operations, including manufacturing procedures
- are highly customer oriented
- have undergone fundamental changes from the traditional production model
- often adopt a lean manufacturing model



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# Principles of Lean Manufacturing

- *Pull Processing* – products are pulled from the consumer end (demand), not pushed from the production end (supply)
- *Perfect Quality* –pull processing requires zero defects in raw material, WIP, and FG inventories
- *Waste Minimization* – activities that do not add value or maximize the use of scarce resources are eliminated
- *Inventory Reduction* – hallmark of lean manufacturing
  - Inventories cost money
  - Inventories can mask production problems
  - Inventories can precipitate overproduction


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# Principles of Lean Manufacturing

- *Production Flexibility* – reduce setup time to a minimum, allowing for a greater diversity of products, without sacrificing efficiency
- *Established Supplier Relations* – late deliveries, defective raw materials, or incorrect orders will shut down production since there are inventory reserves
- *Team Attitude* – each employee must be vigilant of problems that threaten the continuous flow of the production line


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# Lean Manufacturing Model

- Achieve production flexibility by means of:
  - Changes in the physical organization of production facilities
  - Employment of automated technologies
    - CIM, AS/RS, robotics, CAD, and CAM
  - Use of alternative accounting models
    - ABC and value stream accounting
  - Use of advanced information systems
    - MRP, MRPII, ERP, and EDI

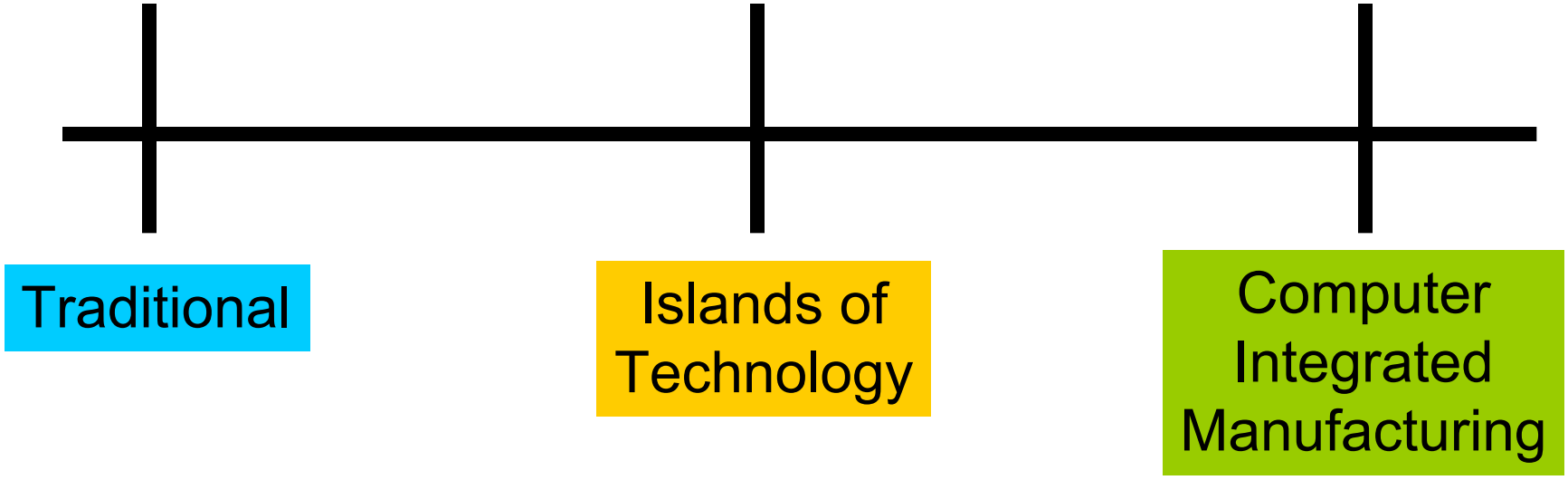

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# Physical Reorganization of the Production Facilities

- Inefficiencies in traditional plant layouts increase handling costs, conversion time, and excess inventories.
- Employees tend to feel ownership over their stations, contrary to the team concept.
- Reorganization is based on *flows through cells* which shorten the physical distance between activities.
  - This reduces setup and processing time, handling costs, and inventories.


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# Progression of Automation in the Manufacturing Process





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# Automating Manufacturing

- **Traditional Approach to Automation**
  - Consists of many different types of machines which require a lot of setup time
  - Machines and operators are organized in functional departments
  - WIP follows a circuitous route through the different operations



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# Automating Manufacturing

- **Islands of Technology**
  - Stand alone islands which employ *computer numerical controlled (CNC)* machines that can perform multiple operations with less human involvement
- **Computer Numerical Controlled (CNC ) Machines**
  - Reduce the complexity of the physical layout
  - Arranged in groups and  in cells to produce an entire part from start to finish
  - Need less set-up time



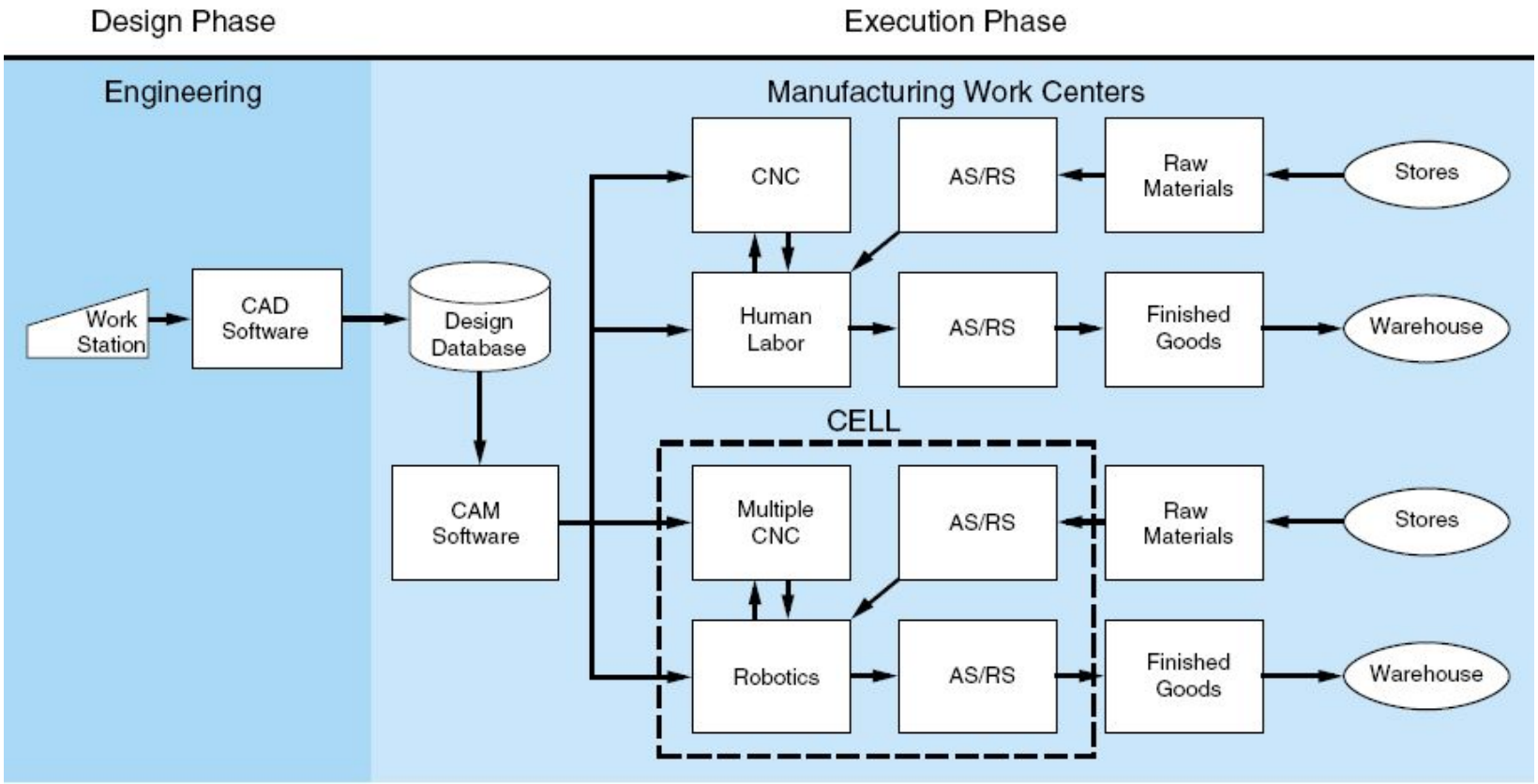
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# Automating Manufacturing

- **Computer Integrated Manufacturing (CIM)**
  - A *completely automated environment* which employs automated storage and retrieval systems (AS/RS) and robotics
- ***Automated Storage and Retrieval Systems (AS/RS)***
  - Replaces traditional forklifts and their human operators with computer-controlled conveyor systems
  - Reduce errors, improved inventory control, and lower storage costs


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# Computer-Integrated Manufacturing (CIM) System





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# Automating Manufacturing

- **Robotics**
  - Use special CNC machines that are useful in performing hazardous, difficult, and monotonous tasks
- **Computer-Aided Design (CAD)**
  - Increases engineers' productivity
  - Improves accuracy
  - Allows firms to be more responsive to market demands
  - Interfaces with CAM and MRPII systems


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# Automating Manufacturing

- **Computer Aided Manufacturing (CAM)**
  - Uses computers to control the physical manufacturing process
  - Provides greater precision, speed, and control than human production processes


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# Achieving World-Class Status

- The world-class firm needs new accounting methods and new information systems that:
  - show what matters to its customers
  - identify profitable products
  - identify profitable customers
  - identify opportunities for improving operations and products
  - encourage the adoption of value-added activities and processes and identify those that do not add value
  - efficiently support multiple users with both financial and nonfinancial information


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# What's Wrong with Traditional Accounting Information?

- *Inaccurate cost allocations* – automation changes the relationship between direct labor, direct materials, and overhead cost
- *Promotes nonlean behavior* – incentives to produce large batches and inventories, and conceal waste in overhead allocations
- *Time lag* – data lag due to assumption that control can be applied after the fact to correct errors
- *Financial orientation* – dollars as the standard unit of measure

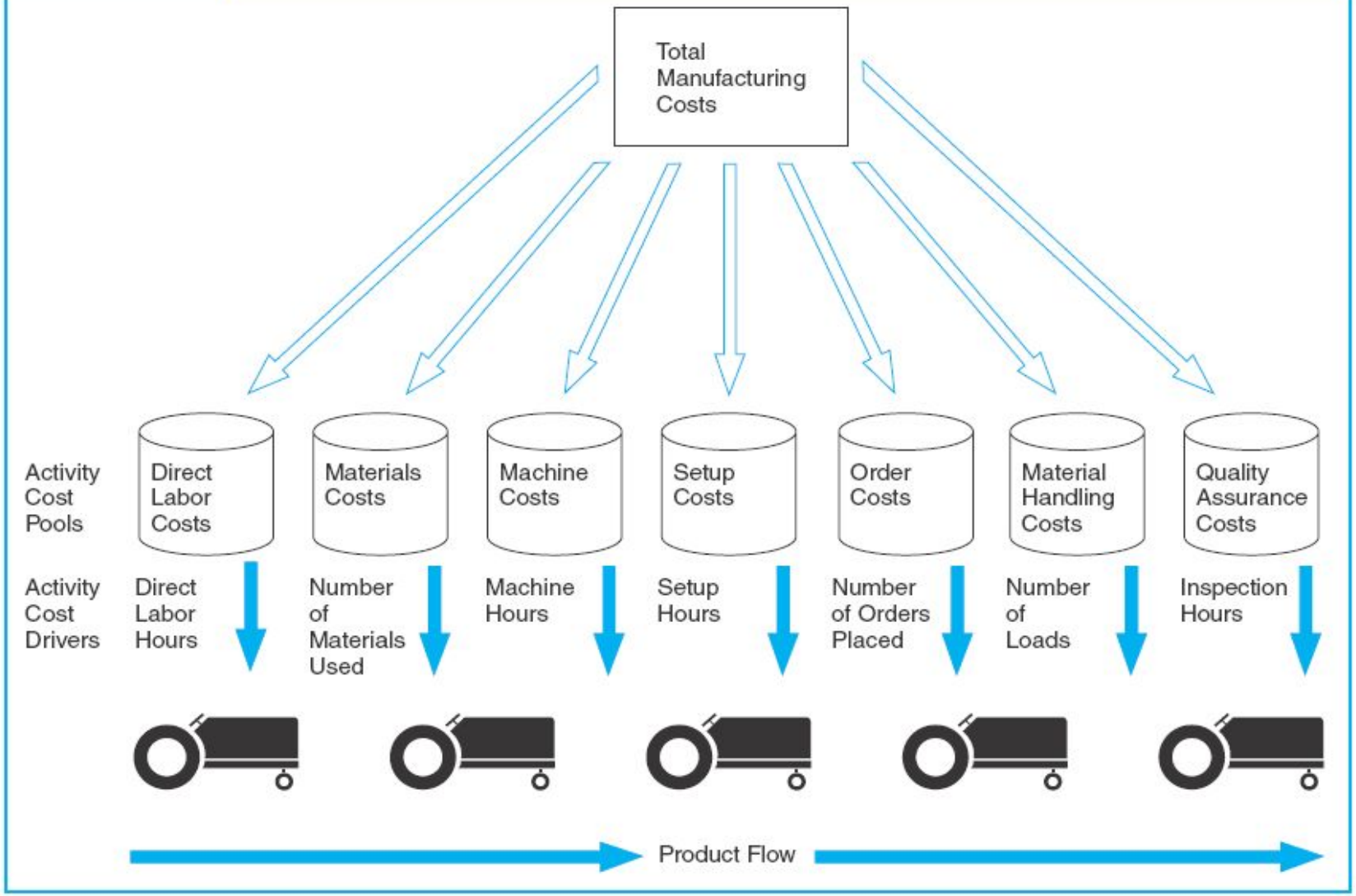

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# Activity Based Costing (ABC)…

- is an information system that provides managers with information about **activities** and **cost objects**
- assumes that activities cause costs and that products (and other cost objects) create a demand for activities
- is different from traditional accounting system since ABC has multiple activity drivers, whereas traditional accounting has only one, e.g. machine hours


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**FIGURE 7-20** Allocation of Overhead Costs to Products under ABC





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# ABC – Pros and Cons

- **Advantages**
  - More accurate costing of products/services, customers, and distribution channels
  - Identifying the most and least profitable products and customers
  - Accurately tracking costs of activities and processes
  - Equipping managers with cost intelligence to drive continuous improvements
  - Facilitating better marketing mix
  - Identifying waste and non-value-added activities
- **Disadvantages**
  - Too time-consuming and complicated to be practical
  - Promotes complex bureaucracies in conflict with lean manufacturing philosophy

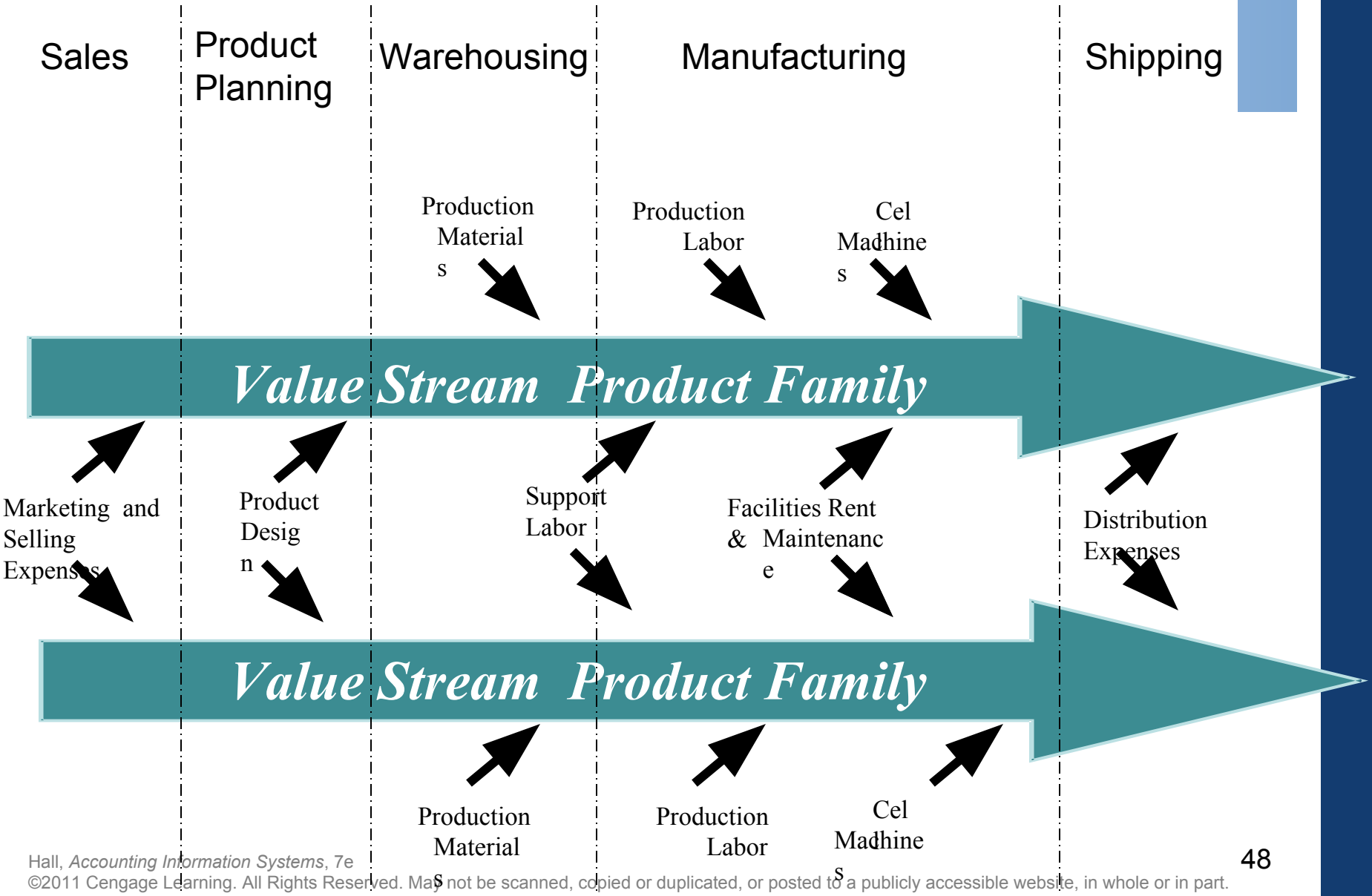

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# Value Stream Accounting

- *Value stream* – all the steps in a process that are essential to producing a product
- Value streams cut across functions and departments
- Captures costs by value stream rather than by department or activity
  - Simpler than ABC accounting
- Makes no distinction between direct and indirect costs
  - Including labor costs


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# Cost Assignment to Value Stream





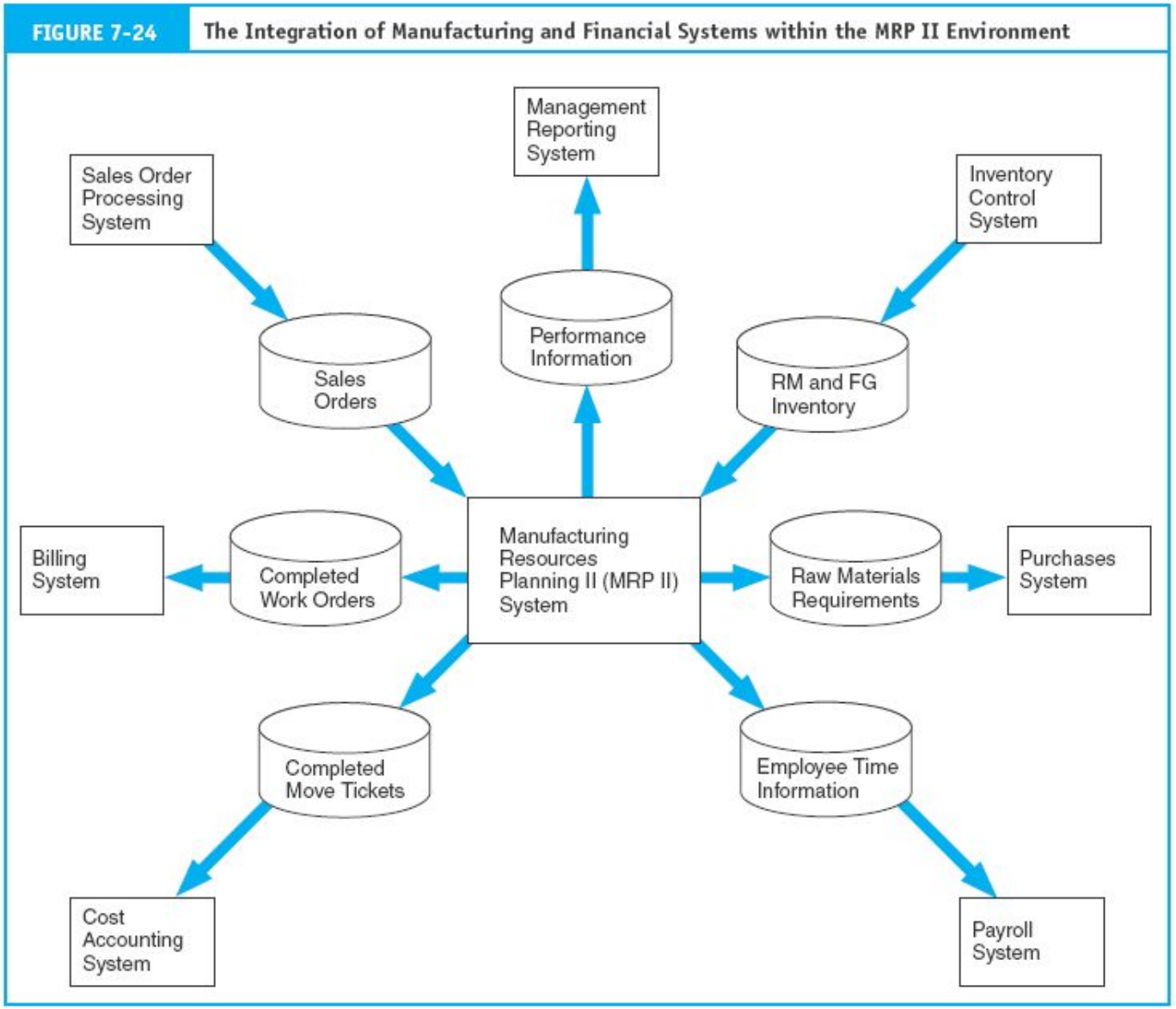
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# Information Systems that Support Lean Manufacturing

- **Manufacturing Resources Planning (MRP)**
  - Ensures adequate raw materials for production process
  - Maintains the lowest possible level of inventory on hand
  - Produce production and purchasing schedules and other information needed to control production
- **MRP II**
  - An extension of MRP
  - More than inventory management and production scheduling – it is a system for coordinating the activities of the entire firm


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**FIGURE 7-24** The Integration of Manufacturing and Financial Systems within the MRP II Environment





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# Information Systems that Support Lean Manufacturing

- **Enterprise Resource Planning (ERP) Systems**
  - Huge commercial software packages that support the information needs of the entire organization, not just the manufacturing functions
  - Automates all business functions along with full financial and managerial reporting capability
- **Electronic Data Interchange (EDI)**
  - External communications with its customers and suppliers via Internet or direct connection


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